Advanced Hive

Friday, 21 August 2026 7:09 PM

Hive. → Is hive a database?

— Hive is not a traditional database.

— Hive is a data warehouse tool/framework that provides a SQL interface on top of Hadoop storage systems.

Q. Does hive store data on it's own?

Actual data is stored in HDFS or cloud based object storage.

Q. What is hive best suited for?

OLAP

Q. Do you prefer hive for real time transactions?

No. It is best suited for batch processing.

* Hive Table Types.

1) Internal Tables: It is managed by hive.

```
Create table employees (
    first_name  STRING,
    Last_name   STRING,
    Phone       Int.
    )
Row Format Delimited
Fields Terminated by ',';
```

2) External Table.

Data already exists outside Hive.

It is not managed by hive, it only stores the metadata.

```
Create External table external_employee(
    first_name  String,
    Last_name   String,
    Phone       Int
    )
Row Format Delimited
Fields Terminated by ','
Location 'user/shubham/folder/external_employee';
```

1

2

path.

* Comparing Internal & External Tables.

Aspect	Managed (Internal Table)	External Table.
Data Ownership	Hive owns metadata & data	Hive only owns metadata
Data location	Stored in hive default warehouse	Stored in a user specified location.
Data lifecycle.	Tightly coupled with the table	Independent of the table.
What happens on DROP TABLE	Data files are deleted	Data files remain intact.
Risk of data loss	High	low
Typical usage	Intermediate datasets	Shared & long term datasets
Control over data	High / Full	low
Multi tool access	Limited (Hive centric)	Easy to share across tools.
Governance.	None for Hive	less.
Best for production data.	Not recommended	Highly Recommended.

* Complex Data Types.

1) Array: It is a collection of values of the same data type.
It is ordered.

```
arr1 = [11, 12, 13, 14, 15]

arr1[0] → 11
```

```
Create table department (
① dept_id  Int,
② teams    Array<String>      -- team names.
)
```

dept_id	teams.
1	"TeamA | TeamB | TeamC"
2	"TeamA | TeamB"
3	"TeamC"

2) Map: Stores data as key-value pair.

```
Create Table department (
dept_id  Int,
→budget  Map<String, Int>   -- budget per subject.
);
```

dept_id | budget.

... 4000, English : 3000"

dept-id	
1	{"Math": 1000, Sci : 7000, ...
2	"Math": 2000, Sci : 8000"
3	⋮

3) Struct: Different data types in a single column.

```
Create table students (
student_id int,
address Struct < street : string , city : string , zip : Int >
);
```

student_id	address
1	{"street": "MG Road", "city": "Blr", "zip": 560001}
2	"
3	"

* Window Functions.

1) Row_number: Assign a sequential numbers

```
Select student, marks,
row_number () over ( order by marks desc) row_num.
```

2) Rank: Assigns ranks to rows.
Skips rank in case of tie.

```
select * from
(Select student_id , marks
rank () over ( order by marks desc) rnk
)
where rnk = 1;
```

3) Dense_rank: Assign rank to rows
Does not skip on ties

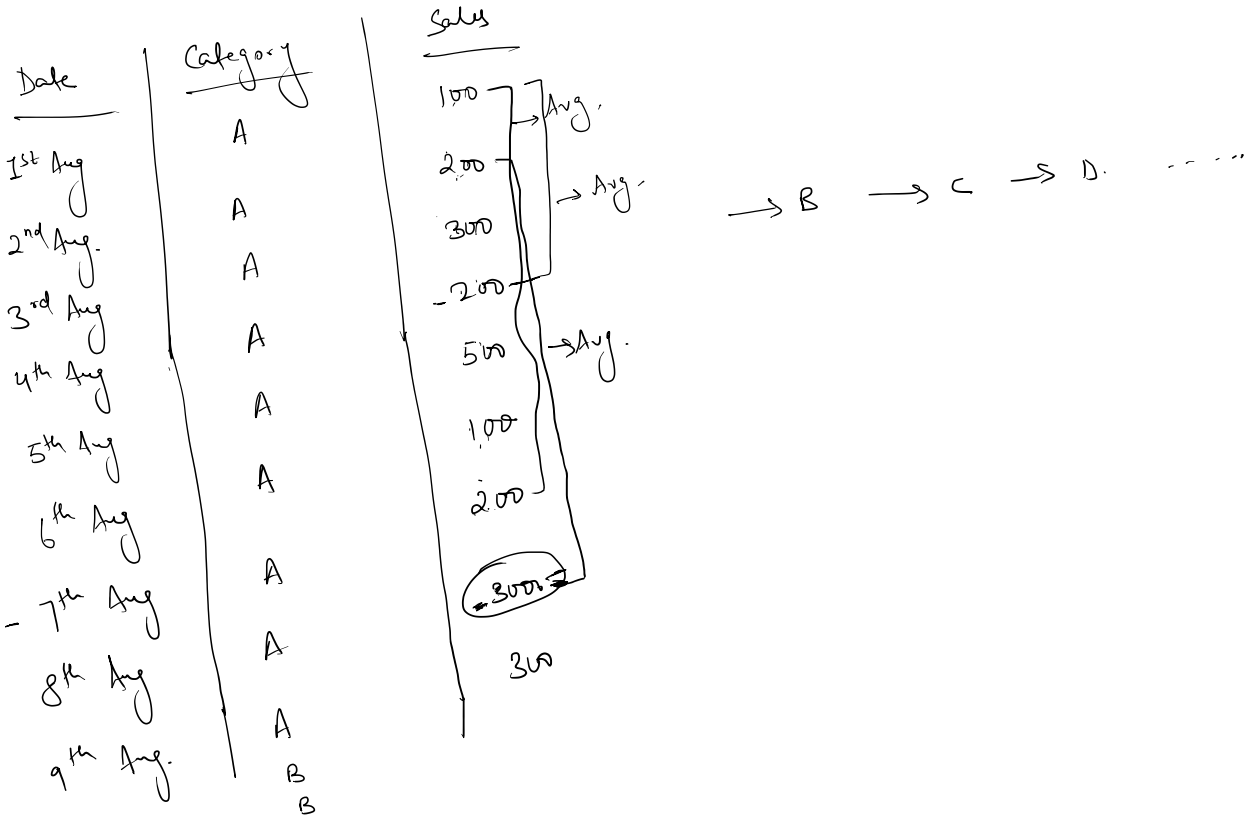
	marks	row_num	rank	dense_rnk.

(rank skips the next rank on tie.)

Student				
A	95 ✓	1	1	1
B	95 ✓	2	1	1
C	90 ✓	3	3	2
D	90	4	3	2
E	85	5	5	3

4) Moving Average:

Q: Find out the last 7 day avg of sales for each category.



Select Date, Category,

avg(Sales) over (partition by category order by date asc.

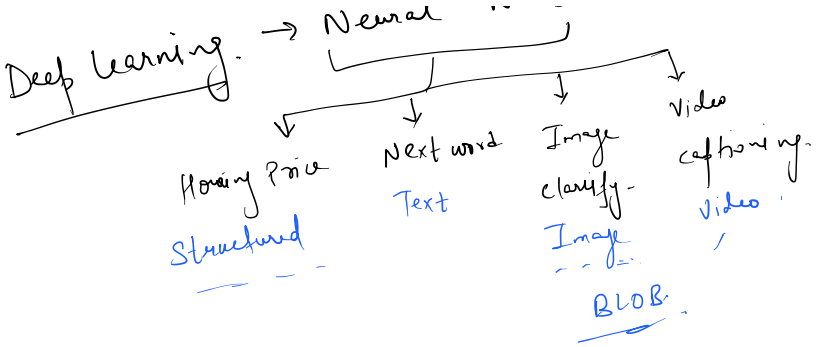
Rows between 6 preceding and Current Row) mvg-avg

6 + 1 = 7.

7 preceding and 1 Preceding.

from

Sales_data.


Deep learning → Neural
Housing Price
Next word
Image classify
Video captioning
Structured
Text
Image
Video
BLOB